

Basic anesthetic monitoring

- Anaesthetist cares for patient throughout anesthesia and checks medications/equipment.
- Continuously evaluate oxygenation, ventilation, circulation, and temperature clinically and by monitors.
- Monitoring begins before induction and continues until full recovery.
- No minor anesthesia.

Cardiovascular monitoring

- Pulse palpation, heart sounds, BP and HR at least every 5 minutes.
- Non-invasive BP: manual or automated.
- Invasive arterial BP: arterial cannula, usually radial, pressure transducer; used for major operations, unstable patients, controlled hypotension, frequent ABG.
- MAP = DBP + 1/3 pulse pressure; complications: thrombosis, infection, bleeding.

ECG and arrest rhythm logic

- ECG evaluates rate, rhythm, myocardial ischemia, electrolytes, and pacemaker.
- ECG does not monitor mechanical myocardial function or cardiac output.
- Shockable rhythms: VF and pulseless VT; non-shockable: asystole and PEA.
- VF is commonest reversible adult cardiac arrest cause; shock shockable rhythms, give adrenaline for non-shockable rhythms.

Ventilation monitoring

- Observe patient color, chest movement, respiratory pattern, reservoir bag movement, and auscultate breath sounds.
- Pulse oximetry continuously monitors arterial O2 saturation and heart rate.
- Normal SpO2 in lecture: 100-97%; later slide lists 95-100%; COPD target may be 88-92%.
- Capnography monitors end-tidal CO2; normal range 35-45 mmHg.

Pulse oximeter errors

- False low: poor perfusion, shock, hypothermia, low BP, vasoconstrictors, motion, nail polish/artificial nails, pigmentation, severe anemia, CO2 retention/acidosis.
- False high: carbon monoxide poisoning; methemoglobinemia often reads around 85%.
- Capnography sudden loss causes: esophageal intubation, obstruction, disconnection, extubation, equipment failure, PE, apnea, arrest.

Temperature and neuromuscular monitoring

- Hypothermia: core temperature <=35 C; common cause of delayed recovery especially infants and elderly.
- Thermistor probes may be nasopharyngeal, esophageal, or rectal.
- Prevent/treat hypothermia: cover exposed surfaces, warm-water/forced-air blanket, warm fluids/blood/irrigation.
- Peripheral nerve stimulator assesses blockade, next relaxant dose, reversal before extubation, and nerve block site.

Fluid status and replacement goals

- Healthy adult balances about 2.5 L water/day/70 kg; adult exchanges 5% TBW/day, infant 15%.
- ECF deficit: dry mouth, thirst, sunken eyes, poor turgor, drowsiness, oliguria, orthostatic hypotension, tachycardia/tachypnea, cool extremities, low CVP, metabolic acidosis.
- ECF expansion: pulmonary edema, dependent edema, S3, raised JVP.
- Goals: intravascular volume, tissue perfusion, organ function, oxygen delivery, avoid hypo/overload, maintain electrolytes and acid-base.

Intraoperative replacement formula

- Total = basal hourly requirement + preoperative deficit + operative loss.
- Basal requirement uses 4-2-1 rule.
- Preoperative deficit = basal hourly requirement x fasting hours; add pathological deficits.
- Operative loss = blood loss + third-space surgical trauma using 4-6-8 ml/kg/hr.
- Guide by BP, HR, urine output 1 ml/kg/hr, CVP 5-10 cm H2O; BP may not fall until about 30% blood volume is lost.
- Replacement rate: first hour 50%, second hour 25%, third hour 25%.

Replacement Fluids And Blood

Type	Use / ratio	Major cautions
Crystalloids	First line; after 20 min about 1/3 stays intravascular; give 3:1 for estimated blood loss.	Normal saline may cause hyponatremia, hyperchloremic acidosis, pulmonary edema. LR is first choice but caution in diabetes, hyperkalemia, renal failure.
5% dextrose	Minimal volume expansion; moves to interstitial/intracellular spaces.	May cause hyperglycemia, hyponatremia; not for resuscitation.
Colloids	Higher osmolality; longer intravascular duration; 1:1 blood loss ratio, max 20 ml/kg/day.	Allergy, no O2 carrying capacity, cross-match interference, renal problems.
Blood/RBCs	Increase O2 capacity. One unit raises Hct about 3% and Hb 1 g/dL.	Transfusion not indicated >10 g/dL; 7-10 depends tolerance/expected loss; <7 indicated.
Reaction management	Stop transfusion, 100% O2, IV fluids, inotropes, steroids, maintain urine output, monitor K/ECG and DIC labs.	Send remaining blood plus patient blood/urine to lab.